KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3371 | 3 Credits: 3 | Program Core | Source-grounded

Introduction to IC Fabrication

- To introduce the fundamentals of IC fabrication process.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3371 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3371.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Integrated Circuit Design and Fabrication
Course code	3371
Course title	Introduction to IC Fabrication
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:3 T:0 P:0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To introduce the fundamentals of IC fabrication process.

Course outcomes

CO	Official outcome	Study level
CO1	10 Understanding preparation and wafer cleaning.	See official syllabus
CO2	Explain the process of epitaxy and oxidation. 11 Understanding	See official syllabus
CO3	Explain lithography and etching process. 10 Understanding Illustrate the process of deposition, diffusion, ion	See official syllabus
CO4	12 Understanding implantation and packaging. Series Exam 2	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	cleaning	4	As specified
Module 2	Explain the process of Epitaxy and Oxidation.	4	As specified
Module 3	Explain Lithography and Etching process.	4	As specified
Module 4	Packaging.	4	As specified

MODULE 1

cleaning

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: cleaning
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

2 Understanding fabrication

2 Understanding fabrication is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding fabrication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding fabrication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding fabrication means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 2 Understanding fabrication [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing

Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the methods of single crystal growth 3 understanding explain chemical-mechanical polishing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding technique

2 Understanding technique is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding technique using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding technique should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding technique means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding technique □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc.

Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. is an official topic in Module 1 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain wafer preparation 3 understanding crystal growth, wafer preparation: introduction & historical perspective- clean room concept- growth of single crystal si-czochralski(cz) and float zone(fz) processes. surface contamination-chemical-mechanical polishing. wafer preparation-di water, chemical cleaning-rca, processing consideration- getting the thermal stress factor etc. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
cleaning
             Explain the concept of clean room in IC
   M1.01                                                         2
Understanding
             fabrication

   M1.02     Explain the methods of single crystal growth        3
Understanding

             Explain chemical-mechanical polishing
   M1.03                                                         2
Understanding
             technique

   M1.04     Explain wafer preparation                           3
Understanding

 Contents:
 Crystal growth, wafer preparation: Introduction & historical perspective- Clean
room
 concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes.
Surface
 Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical
 cleaning-RCA, processing consideration- getting the thermal stress factor etc.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the process of Epitaxy and Oxidation.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the process of Epitaxy and Oxidation.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth

Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the process of epitaxy 3 understanding illustrate epitaxial valuation and growth should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding mechanism

3 Understanding mechanism is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding mechanism means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding mechanism □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Outline the process of oxidation

Outline the process of oxidation is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the process of oxidation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the process of oxidation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the process of oxidation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Outline the process of oxidation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove’s Model-Oxidation system.

Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove’s Model-Oxidation system. is an official topic in Module 2 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove’s Model-Oxidation system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the kinetics of oxidation 2 understanding epitaxy and oxidation: epitaxy process-vapor phase epitaxy, molecular beam epitaxy. epitaxial valuation-growth mechanism and kinetics. oxidation techniques-dry and wet oxidation. device isolation-locos-kinetics of oxidation. deal grove’s model-oxidation system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove’s Model-Oxidation system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove’s Model-Oxidation system. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the process of Epitaxy and Oxidation.

   M2.01     Explain the process of Epitaxy                       3
Understanding
             Illustrate Epitaxial valuation and growth
   M2.02                                                          3
Understanding
             mechanism
   M2.03     Outline the process of oxidation                     3
Understanding

   M2.04     Describe the kinetics of oxidation                   2
Understanding

             Series Test-1                                        1
Contents:
Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy.
Epitaxial Valuation-Growth mechanism and kinetics.
Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of
Oxidation.
Deal Grove's Model-Oxidation system.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain Lithography and Etching process.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain Lithography and Etching process.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate the concept of Lithography

Illustrate the concept of Lithography is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the concept of Lithography using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the concept of lithography should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the concept of Lithography means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Illustrate the concept of Lithography [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain different types of Lithographic processes

Explain different types of Lithographic processes is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different types of Lithographic processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different types of lithographic processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different types of Lithographic processes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain different types of Lithographic processes [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the concept of etching

Explain the concept of etching is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of etching using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of etching should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of etching means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain the concept of etching [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and

Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and is an official topic in Module 3 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the different types of etching process 3 understanding lithography & etching: lithography-overview of lithography-radiation sources-masks-photoresist-components of photoresist- optical aligners-resolution-depth of focus-advanced lithography-electron beam lithography-x-ray lithography-ion lithography. etching-anisotropy-selectivity-wet etching, plasma etching, plasma properties-reactive - ion etching. illustrate the process of deposition, diffusion, ion implantation and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain Lithography and Etching process.

  M3.01     Illustrate the concept of Lithography                2
Understanding

  M3.02     Explain different types of Lithographic processes    4
Understanding

  M3.03     Explain the concept of etching                       1
Understanding

  M3.04     Explain the different types of etching process       3
Understanding

Contents:
Lithography & Etching:
Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components
of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-
Electron
beam Lithography-X-ray Lithography-Ion Lithography.
Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-
Reactive -
Ion Etching.

            Illustrate the process of Deposition, Diffusion, Ion implantation and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Packaging.

This module is presented from the official syllabus scope for Introduction to IC Fabrication. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Packaging.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain Diffusion mechanism

Explain Diffusion mechanism is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Diffusion mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain diffusion mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Diffusion mechanism means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain Diffusion mechanism [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain various ion implantation techniques

Explain various ion implantation techniques is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various ion implantation techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various ion implantation techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various ion implantation techniques means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain various ion implantation techniques □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Explain different deposition mechanisms

Explain different deposition mechanisms is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different deposition mechanisms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different deposition mechanisms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different deposition mechanisms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain different deposition mechanisms [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process-Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in

Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process-Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, "VLSI Technology", McGraw Hill Publication, 2003. T2 S.K Gandhi, "VLSI Fabrication Principle", Willy-India Pvt Ltd, 2008. T3 Suresh Babu "Solid State Device & Technology", Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modelling", Pearson Education Publication, 2009. R2 Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in is an official topic in Module 4 of Introduction to IC Fabrication. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain metallization and packaging 3 understanding deposition, diffusion, ion implantation and packaging: model of diffusion in solids-atomic diffusion mechanism-measurement technique-range theory-implantation equipment-annealing shallow junction-high energy implantation-deposition process-chemical vapor deposition-physical vapor deposition-patterning-metallization and packaging-baked ic design. text / reference: t/r book title/author t1 s.m sze, “vlsi technology”, mcgraw hill publication, 2003. t2 s.k gandhi, “vlsi fabrication principle”, willy-india pvt ltd, 2008. t3 suresh babu “solid state device & technology”, sanguine, 2005. r1 j.d plummer, m.d deal and peter b. griffin, “silicon vlsi technology: fundamentals, practice and modelling”, pearson education publication, 2009. r2 stephen a. campbell, “fabrication engineering at the micro and nano scale”, oxford university press, 2013 online resources: sl. no. website link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-

Aspect	What to prepare
	Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Packaging.

  M4.01     Explain Diffusion mechanism                           3
Understanding

  M4.02     Explain various ion implantation techniques           3
Understanding

  M4.03     Explain different deposition mechanisms               3
Understanding

  M4.04     Explain metallization and packaging                   3
Understanding

            Series Test-II                                        1

Contents:
Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in
solids-
Atomic diffusion mechanism-Measurement technique-Range theory-Implantation
equipment-Annealing shallow junction-High energy implantation-Deposition process-
 Chemical vapor deposition-Physical          vapor    deposition-Patterning-
Metallization   and
 Packaging-Baked IC design.

Text / Reference:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Understanding fabrication. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding fabrication*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the methods of single crystal growth 3 Understanding Explain chemical-mechanical polishing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding technique. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding technique*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain wafer preparation 3 Understanding Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress* *factor etc.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding mechanism. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding mechanism*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline the process of oxidation. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the process of oxidation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system.. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the kinetics of oxidation 2 Understanding Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics. Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation. Deal Grove's Model-Oxidation system.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Illustrate the concept of Lithography. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the concept of Lithography*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain different types of Lithographic processes. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different types of Lithographic processes*. State the governing principle, list the key components or stages, and close with one relevant application or observation.

stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the concept of etching. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the concept of etching*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the different types of etching process 3 Understanding Lithography & Etching: Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography. Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching. Illustrate the process of Deposition, Diffusion, Ion implantation and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain Diffusion mechanism. (3 marks)

Answer: Begin with the standard definition or identity of *Explain Diffusion mechanism*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain various ion implantation techniques. (3 marks)

Answer: Begin with the standard definition or identity of *Explain various ion implantation techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain different deposition mechanisms. (3 marks)

Answer: Begin with the standard definition or identity of *Explain different deposition mechanisms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-

Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in. (3 marks)

Answer: Begin with the standard definition or identity of *Explain metallization and packaging 3 Understanding Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids- Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process- Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design. Text / Reference: T/R Book Title/Author T1 S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003. T2 S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008. T3 Suresh Babu “Solid State Device & Technology”, Sanguine, 2005. R1 J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009. R2 Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013 Online resources: Sl. No. Website Link 1. http://www.asic-world.com 2. https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech 3. https://www.testbench.in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding fabrication**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain the process of Epitaxy 3 Understanding Illustrate Epitaxial valuation and growth**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the concept of Lithography**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain Diffusion mechanism**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link http://www.asic-world.com
- https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech https://www.testbench.in

IN-PAGE SEARCH

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